

OBJECTIVE

Software Engineering graduate with research experience in computer vision and deep learning. Interested in efficient deep learning — especially making vision transformers faster and lighter through token compression, knowledge distillation, and efficient architecture design, while keeping them accurate. Eager to contribute to research that makes large vision models practical for real-world and resource-limited settings.

EDUCATION

Fall 2022 – Spring 2026 **B.S. Software Engineering — Gold Medalist, CGPA 3.92/4.00**
University of Haripur, KP, Pakistan

RESEARCH INTERESTS

Efficient Deep Learning · Self-Supervised Representation Learning · Foundation Models · Image Recognition & Object Detection · Knowledge Distillation · Vision-Language Models

MAJOR RESEARCH PROJECT

2025 – 2026 **Augmented Reality & Deep Learning Assisted Enhanced Museum Experience**
Final Year Research Project (Thesis) — University of Haripur

- Built a multimodal AI system integrating computer vision, deep learning, speech interaction, and multilingual assistance for real-world cultural heritage recognition (case study: Taxila Museum).
- Curated a custom dataset of Taxila Museum artifacts; performed end-to-end data collection, preprocessing, and augmentation pipelines; developed vision-language recognition pipelines for low-resource artifact categories.
- Research paper on hybrid vision-language recognition for cultural heritage systems under preparation for publication.
- Nominated by the Directorate General of Science & Technology (DoST), KP for research funding, recognizing scientific merit and societal impact.
- Awarded Top Project at the Final Year Project Exhibition, University of Haripur.

ACADEMIC PROJECTS

Explainable AI for Knee Osteoarthritis Detection & Severity Grading — Python, TensorFlow, DenseNet-121

- DenseNet-121 ensemble for 5-grade knee osteoarthritis severity classification with Grad-CAM interpretability heatmaps.

PulmoVision: Automated Pneumonia Diagnosis Using Deep Learning — Python, TensorFlow, ResNet

- Developed a ResNet-based transfer learning pipeline for automated pneumonia detection and classification from chest X-ray images.

Performance Benchmarking of Machine Learning Algorithms — Python, scikit-learn, Pandas

- Benchmarked multiple machine learning algorithms across datasets and problem instances, analyzing performance variations to identify relationships between dataset characteristics and algorithm effectiveness.

AI-Powered Nail Condition Analyzer — Python, TensorFlow, OpenCV

- CNN-based diagnostic assistant for medical image classification of nail conditions with preprocessing pipelines.

Encoder-Aware Visual Token Compression for Efficient VLMs — Python, PyTorch, ViT

- Implementing adaptive, encoder-dependent visual token compression for vision-language models, benchmarking a random baseline and Top-K pruning against an adaptive compression method on ViT.

Data Structures & Algorithms Visualization Suite — C++

- Desktop application with graphical visualization of sorting, tree traversals, and graph pathfinding algorithms.

TECHNICAL SKILLS

Languages	Python, JavaScript, C++, C#
Frameworks	Hugging Face Transformers, PyTorch, TensorFlow, OpenCV, Scikit-learn, Streamlit
Dev Tools	Git/GitHub, Linux/Bash, REST APIs, Flask, Streamlit

EXPERIENCE

- Jul 2026 – Present **AI Intern**
UAVs Lab, National Radio and Telecommunication Corporation (NRTC) — Haripur, Pakistan
- Working on an object detection and pattern recognition module for a UAV-based mission: comparing before/after aerial imagery from a drone's outbound and return flight path to identify and flag missing objects.
 - Collaborating within a larger multi-module team project, owning the detection and comparison component end-to-end.
- Jul 2025 – Aug 2025 **IT Directorate Intern**
Senate of Pakistan — Islamabad, Pakistan
- Contributed to software development and web-based systems within the IT Directorate.
 - Assisted in maintaining and troubleshooting internal web applications used across departments.
 - Collaborated with senior IT staff on requirement gathering and testing for ongoing system upgrades.
- Jul 2024 – Aug 2024 **Game Development Intern**
Codematics Inc. — Abbottabad, Pakistan
- Developed 2D/3D Unity projects, including animations and shader graphs.
 - Explored AI-driven gameplay elements and prototyped interactive mechanics.
 - Collaborated with a small team on game design iteration and playtesting feedback.

AWARDS & ACHIEVEMENTS

Gold Medal, BS Software Engineering (CGPA 3.92/4.00), University of Haripur
Top Project, Final Year Project Exhibition, University of Haripur
PM Youth Laptop Scheme — awarded for academic excellence
A1 grades in SSC & FSc (2019–2021); Top 10, Abbottabad Board (FSc); 93rd percentile, Engineering/IT entry test
Top Performer, Python Frenzy Competition, IT Club
Award for Organizing UoH 4th Convocation (2025)
Participant, 1st International Conference on Computational Sciences

CERTIFICATIONS

- AI in Healthcare Specialization — Stanford University
- Deep Learning for Computer Vision: Techniques and Applications — Khalifa University
- Computer Vision Specialization — University of Colorado Boulder
- Modern AI Models for Vision and Multimodal Understanding — University of Colorado Boulder
- Multimodal Intelligence: Vision, Audio and Language in Action — Coursera
- Build Multimodal Generative AI Applications — IBM
- Fundamentals of Building AI Agents — IBM
- Explainable AI: Scene Classification and Grad-CAM Visualization — Coursera
- Interpretable Machine Learning — Duke University
- Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI
- AI in Healthcare Webinar — UAEU
- Crash Course on Python — Google

LEADERSHIP & SERVICE

- Jul 2025 – Present Vice President, IT Computing Society, University of Haripur
- Dec 2025 Career Expo 2025 Organizer, University of Haripur
- May 2025 Event Manager, IT Department Orientation, University of Haripur
- Jan 2024 Team Lead, Social Work Visit — Aghosh AI-Khidmat, Haripur

REFERENCES

Available upon request.